

# Resonant Non-Gaussianity from Non-Bunch-Davies Vacua

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## Abstract

We study the impact of non-Bunch-Davies vacua on inflationary correlators within a Hadamard-preserving regularization. Using an exact contour representation, the regulated two-point kernel cleanly separates the ultraviolet Hadamard sector from oscillatory contributions associated with the excited initial state. The resulting propagator is consistently inserted into the Schwinger-Keldysh expansion to compute a tree-level bispectrum sourced by cubic interactions.

The signal remains slow-roll suppressed but departs from the standard single-field bispectrum through oscillatory contributions generated by interference between positive- and negative-frequency sectors. In contrast with phenomenological approaches to resonant non-Gaussianity, the present oscillatory structure emerges directly from the analytic properties of contour-deformed non-Bunch-Davies propagators.

The squeezed-limit behavior departs from the standard single-field Maldacena scaling in the sense that the leading infrared enhancement is absent and the bispectrum remains finite. This reflects sensitivity to the excited non-Bunch-Davies initial state rather than a breakdown of attractor single-field inflation. For sub-Hubble scales, the oscillatory sector admits an effective short-distance kernel interpretation, and the bispectrum is predominantly supported in equilateral configurations.

The framework provides a controlled separation between ultraviolet Hadamard structure and state-dependent oscillatory effects, yielding a propagator-based realization of resonant non-Gaussian signatures associated with non-Bunch-Davies initial conditions.

**Keywords:** inflation, non-Gaussianity, non-Bunch-Davies vacua, in-in formalism, Hadamard regularization, primordial bispectrum

# 1 Introduction

Non-Gaussian features of single-field inflation have been extensively studied in the context of inflationary quantum fluctuations using the “In-in”, or Schwinger–Keldysh, formalism [1–6]. In this framework, cosmological observables such as the power spectrum and the bispectrum are obtained from expectation values of interacting quantum fields evolving in a quasi-de Sitter background.

A central assumption in most standard treatments is the choice of the Bunch–Davies vacuum, implemented through Hankel-function mode functions and the corresponding adiabatic early-time behavior. In the present work, we instead consider a non-Bunch–Davies initial state constructed from a Bessel-function mode basis. This choice introduces a nontrivial mixing of positive- and negative-frequency components and consequently modifies the analytic structure of the in-in propagator. In particular, it leads to oscillatory contributions in correlation functions associated with the phase structure of the excited state. Oscillatory non-Gaussianities of this type have been extensively studied in the context of resonant and excited-state inflationary models [7–10], where interference effects in the in-in formalism generate characteristic momentum-dependent phase structures in the bispectrum.

Because inflationary perturbations are primarily constrained through their large-scale behavior, most renormalization schemes for cosmological correlators have focused on preserving the infrared structure of the theory. In particular, adiabatic regularization and related subtraction methods have been extensively employed to control ultraviolet divergences while maintaining the nearly scale-invariant behavior of the power spectrum [11–14]. However, the short-distance structure of cosmological correlators remains of independent conceptual interest, especially in relation to the local Hadamard singularity required by quantum field theory in curved spacetime. This motivates the study of alternative regularization procedures capable of isolating ultraviolet contributions while preserving the local causal structure of the propagator.

Motivated by these considerations, the present work revisits the exact-solution framework of cosmological perturbation theory and analyzes the two-point correlator of the Mukhanov–Sasaki (MS) variable directly from the integral representation of its mode solutions. By exploiting the analytic structure associated with the non-Bunch–Davies Bessel basis, the Fourier representation of the correlator can be reorganized in terms of special functions whose singular structure becomes explicitly tractable. This allows the coincidence-limit behavior of the correlator to be isolated analytically and yields a Hadamard-preserving regularized propagator at tree level.

The momentum-space propagator obtained in this way should be understood as an intermediate regulated kernel entering the Schwinger–Keldysh expansion of higher-order correlators, rather than as a direct definition of a physical observable spectrum. Its role is to provide a controlled short-distance representation of the two-point function that can be consistently inserted into tree-level in-in diagrams. In particular, the contour deformation isolates the ultraviolet Hadamard contribution while retaining the oscillatory phase structure induced by the non-Bunch–Davies initial state.

The resulting kernel is then employed in the computation of the bispectrum within the in-in formalism. In contrast with standard infrared-oriented analyses, the present construction isolates the short-distance sector of the correlator while retaining explicit

dependence on the state-induced phase structure. This leads to oscillatory contributions in momentum space originating from interference effects associated with the non-Bunch-Davies vacuum.

This representation therefore provides an alternative analytic continuation framework in which the ultraviolet structure of the propagator can be separated from its state-dependent oscillatory sector in a controlled manner, allowing the latter to be propagated consistently into higher-order correlation functions.

The present paper is organized as follows. In Sec. II, we introduce a Hadamard-preserving regularization procedure based on an exact integral identity and apply it to the two-point correlator of the MS variable in position space. The method performs an analytic continuation with respect to the radial coordinate and organizes the coincidence-limit behavior of the correlator in a form consistent with the expected ultraviolet Hadamard structure.

In Sec. III, we extend the analysis to momentum space and introduce the Euclideanized spectral representation of the propagator. We clarify its relation to zeta-function and heat-kernel methods and show how the oscillatory contributions arise from the non-Bunch-Davies structure of the mode basis.

Finally, in Sec. IV, we compute the bispectrum using the Hadamard-regularized propagators in momentum space under controlled short-distance approximations. We discuss the resulting oscillatory structure, its dependence on the initial state, and its implications for the shape dependence of primordial non-Gaussianities.

## 1.1 Mukhanov-Sasaki in $(n + 1)$ -dimension

In concordance with dimensional regularization, we present the equations of motion of the MS field  $v$  in a flat FLRW  $(n + 1)$ -spacetime. In momentum space, the MS equation can be written as the equation of motion of a harmonic oscillator with a time-dependent frequency,

$$v_{\mathbf{k}}'' + \left( k^2 - \frac{z''}{z} \right) v_{\mathbf{k}} = 0, \quad z := a^{(n-1)/2} \sqrt{2\varepsilon}. \quad (1)$$

Furthermore, in the slow-roll approximation the time-dependent frequency can be expressed as

$$\frac{z''}{z} = \frac{1}{\tau^2} \left( \nu^2 - \frac{1}{4} \right) + \mathcal{O}(\varepsilon^2), \quad \nu := \frac{n}{2} + \frac{n+1}{2} \varepsilon - \delta. \quad (2)$$

In the standard treatment of inflationary perturbations (Refs. [15–17]) the mode functions are usually written in terms of Hankel functions. This condition corresponds to the Bunch-Davies vacuum (see Eq. (2.14) of Ref. [18]), which satisfies

$$v_{\mathbf{k}} \rightarrow \frac{e^{-ik\tau}}{\sqrt{2k}}, \quad \tau \rightarrow -\infty. \quad (3)$$

In consequence, the Hankel solution in  $n$ -spatial dimensions is written as

$$v_{\mathbf{k}} = \frac{\sqrt{\pi}}{2} e^{i(\nu+\frac{1}{2})\frac{\pi}{2}} \sqrt{-\tau} H_{\nu}^{(1)}(-k\tau). \quad (4)$$

Additionally, the comoving curvature perturbation is the physical field introduced to compute cosmological observables

$$\mathcal{R} := z^{-1}v, \quad z := a^{(n-1)/2} \sqrt{2\varepsilon}. \quad (5)$$

## 2 Hadamard-based regularization

In the Schwinger–Keldysh formalism, ultraviolet singularities are commonly regulated through an  $i\epsilon$  prescription implemented at the level of the invariant distance,

$$y(x, x') = a(\tau_1)a(\tau_2)H^2\Delta x^2, \quad (6a)$$

$$\Delta x^2 = |\mathbf{x} - \mathbf{x}'|^2 - (\Delta\tau + i\epsilon)^2, \quad (6b)$$

which guarantees the correct causal ordering of fields inside perturbative diagrams. Such prescriptions are standard in loop computations in de Sitter space [19].

In the present work, however, we consider the integral representation of the Wightman function obtained directly from the mode expansion of the MS field. In this representation, the ultraviolet divergence originates from the spatial Fourier transform of the mode functions rather than from time integrations in loop diagrams.

To regulate this short-distance singularity, we introduce an analytic continuation of the radial separation appearing in the Fourier kernel. The continuation acts only on the integration contour and leaves the arguments of the Bessel mode functions unchanged, which is technically convenient in quasi-de Sitter backgrounds where slow-roll corrections are encoded in the mode structure itself.

The resulting contour-deformed propagator preserves the local Hadamard singular structure while providing a regulated momentum-space kernel that can be consistently inserted into higher-order Schwinger–Keldysh correlators. In this sense, the analytic continuation introduced here should be understood as an intermediate regularization procedure for the computation of interacting diagrams rather than as a modification of the physical observables themselves.

Finally, expressing the mode functions in terms of Bessel functions instead of Hankel functions corresponds to working with a linear combination of positive- and negative-frequency modes. The resulting propagator therefore defines a non-Bunch–Davies spectral representation, whose oscillatory structure generates the ultraviolet contributions studied in the bispectrum calculation below.

Let us begin by taking an alternative solution to the MS equation (1),

$$v_{\mathbf{k}}(\tau) = \frac{\sqrt{\pi}}{2} \sqrt{-\tau} J_{\nu}(-k\tau), \quad \nu = \frac{n}{2} + \frac{n+1}{2} \varepsilon - \delta, \quad (7)$$

where the constant of normalization is chosen accordingly to the asymptotic expansion

$$J_\nu(\rho) \rightarrow \sqrt{\frac{2}{\pi\rho}} \cos\left(\rho - \nu\frac{\pi}{2} - \frac{\pi}{4}\right), \quad \rho \rightarrow \infty, \quad (8)$$

along with the vacuum prescription

$$v_{\mathbf{k}} \rightarrow \frac{1}{\sqrt{2k}} \cos(-k\tau), \quad \tau \rightarrow -\infty. \quad (9)$$

As we can see, this definition is different from Eq. (3)<sup>1</sup>.

With this new solution the correlator turns out to be

$$\begin{aligned} \langle \hat{v}(x) \hat{v}(y) \rangle &= \int \frac{d^n \mathbf{k}}{(2\pi)^n} e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{y})} v_{\mathbf{k}}(\tau_1) v_{\mathbf{k}}^*(\tau_2) \\ &\approx \Omega_n \frac{\pi \sqrt{\tau_1 \tau_2}}{4(2\pi)^n} \int_0^\infty dk k^{n-1} e^{ikr} J_\nu(-k\tau_1) J_\nu(-k\tau_2), \quad r := |\mathbf{x} - \mathbf{y}|, \end{aligned} \quad (11)$$

where an approximation has been applied to neglect the direction of the momentum vectors.

This integral (11) can be calculated by using an old result (Ref. [20]) from the Theory of Bessel functions by Beltrami, Sommerfeld, Gengenbauer and Macdonald (BSGM),

$$\int_0^\infty dk k^{\mu-1} e^{-ak} J_\nu(bk) J_\nu(c k) = \frac{(bc)^\nu \Gamma(\mu + 2\nu)}{\pi a^{\mu+2\nu} \Gamma(2\nu + 1)} I(\mu, \nu, z), \quad (12a)$$

$$I(\mu, \nu, z) := \int_0^\pi {}_2F_1\left(\frac{\mu + 2\nu}{2}, \frac{\mu + 2\nu + 1}{2}; \nu + 1; z\right) \sin^{2\nu}(\phi) d\phi, \quad (12b)$$

$$z(\phi) := -\frac{b^2}{a^2} - \frac{c^2}{a^2} + 2\frac{bc}{a^2} \cos(\phi), \quad (12c)$$

$$\Re(a \pm ib \pm ic) > 0, \quad \Re(\mu + 2\nu) > 0.$$

By comparison between Eqs. (11) and (12) it can be noticed that to fulfilled the restriction of the domain<sup>2</sup> an analytic continuation must be performed in one of the variables:  $r$ ,  $\tau_1$  or  $\tau_2$ , in order to have a non-vanishing real part<sup>3</sup>  $\Re(a \pm ib \pm ic) > 0$ , where we have defined  $a := -ir$ ,  $b := -\tau_1$ ,  $c := -\tau_2$ .

<sup>1</sup>Also, by noticing that

$$\cos(-k\tau) = \frac{e^{ik\tau} + e^{-ik\tau}}{2}, \quad (10)$$

is clear that we are mixing incoming with outgoing waves, or equivalently negative with positive frequency modes.

<sup>2</sup>The restriction  $\Re(\mu) > -\frac{1}{2}$  is trivial because  $\mu, \nu$  are always positive.

<sup>3</sup>In this case, the four equations are the same because  $b$  and  $c$  are real, but in general one would have to verify the four equations individually.

In order to preserve the structure of the Bessel mode functions, let us map  $r$  to the complex plane by making  $r \rightarrow r + i\delta r$ . We have that

$$\Re(a \pm ib \pm ic) = \delta r > 0,$$

is satisfied, provided that  $\delta r > 0$ .

The computation of the last integral, Eq. (12b), is done in the Appendix (A). By inserting the result of Eq. (A7) into the BSGM integral (12), the correlator can be approximated as

$$\begin{aligned} \langle \hat{v}(x) \hat{v}(y) \rangle &\approx \frac{\Omega_n}{8(2\pi)^n} \frac{\Gamma(\mu + 2\nu)}{\Gamma(2\nu + 1)} \\ &\left\{ \frac{(\nu)_2 (\tau_1 \tau_2)^{\nu - \frac{3}{2}}}{(\frac{\mu+2\nu-2}{2})_2 (\frac{\mu+2\nu-1}{2})_2} (-ir + \delta r)^{4-\mu-2\nu} \right. \\ &\quad \left[ {}_2F_1 \left( \frac{\mu + 2\nu - 4}{2}, \frac{\mu + 2\nu - 3}{2}; \nu - 1; -\frac{(\tau_1 + \tau_2)^2}{(-ir + \delta r)^2} \right) \right. \\ &\quad \left. + {}_2F_1 \left( \frac{\mu + 2\nu - 4}{2}, \frac{\mu + 2\nu - 3}{2}; \nu - 1; -\frac{(\tau_1 - \tau_2)^2}{(-ir + \delta r)^2} \right) \right] \\ &\quad + \frac{(\nu)_3 (\tau_1 \tau_2)^{\nu - \frac{5}{2}}}{2(\frac{\mu+2\nu-2}{2})_3 (\frac{\mu+2\nu-1}{2})_3} (-ir + \delta r)^{6-\mu-2\nu} \\ &\quad \left[ {}_2F_1 \left( \frac{\mu + 2\nu - 6}{2}, \frac{\mu + 2\nu - 5}{2}; \nu - 2; -\frac{(\tau_1 + \tau_2)^2}{(-ir + \delta r)^2} \right) \right. \\ &\quad \left. - {}_2F_1 \left( \frac{\mu + 2\nu - 6}{2}, \frac{\mu + 2\nu - 5}{2}; \nu - 2; -\frac{(\tau_1 - \tau_2)^2}{(-ir + \delta r)^2} \right) \right] \Bigg\}, \end{aligned} \tag{13}$$

when  $\sin^{2\nu}(\phi) \approx \sin^3(\phi)$  and where  $(\alpha)_\ell := \alpha(\alpha - 1) \cdots (\alpha - \ell + 1)$  is the descending factorial symbol.

At this stage, the expression (13) takes a form closely related to the seminal result obtained by Bunch & Davies, Eq. (2.11) of Ref. [18]. This agreement provides a useful consistency check of the present approach. The remaining differences arise from the particular representation of a Non-Bunch-Davies vacua employed here.

## 2.1 Truncation to leading order

The hypergeometric representation of Eq. (13) provides an exact expression for the correlator. However, in order to analyze the short-distance behavior relevant for regularization at tree-level, we consider the limit  $\delta r \rightarrow 0$ , which controls the coincidence

structure of the two-point function. In this limit the asymptotic expansion of the hypergeometric function reproduces the leading Hadamard singularity. Higher-order terms remain regular and therefore do not affect the singular structure of the correlator. Consequently, the truncation of the series isolates the leading Hadamard contribution that governs the ultraviolet behavior of the two-point function at tree-level.

According to Eq. (13), we can define the regularized correlator as

$$\langle \hat{v}(x)\hat{v}(y) \rangle_{\text{reg}} := \sum_{\ell, \mathfrak{s}=1}^2 e_{(\ell, \mathfrak{s})} A^{(\ell)} (-ir)^{\alpha(\ell)} f\left(\mathbf{v}^{(\ell)}; c^{(\mathfrak{s})} r^{-2}\right), \quad (14a)$$

$$\left(e_{(\ell, \mathfrak{s})}\right)_{1 \leq \ell, \mathfrak{s} \leq 2} := \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad (14b)$$

$$A^{(\ell)} := \frac{\Omega_n}{8(2\pi)^n} \frac{\Gamma(\mu + 2\nu)}{\Gamma(2\nu + 1)} \frac{(\nu)_{\ell+1}}{2^{\ell-1} \left(\frac{\mu+2\nu-2}{2}\right)_{\ell+1} \left(\frac{\mu+2\nu-1}{2}\right)_{\ell+1}} (\tau_1 \tau_2)^{\nu - \frac{1}{2} - \ell},$$

$$\ell = 1, 2, \quad (14c)$$

$$\alpha(\ell) := 2 + 2\ell - \mu - 2\nu, \quad \ell = 1, 2, \quad (14d)$$

$$\mathbf{v}^{(\ell)} := \left( \frac{\mu + 2\nu - 2 - 2\ell}{2}, \frac{\mu + 2\nu - 1 - 2\ell}{2}, \nu - \ell \right), \quad \ell = 1, 2, \quad (14e)$$

$$c^{(\mathfrak{s})} := \left( \tau_1 (-)^{\mathfrak{s}+1} \tau_2 \right)^2, \quad \mathfrak{s} = 1, 2, \quad (14f)$$

by taking the limit  $\delta r \rightarrow 0$ .

For the following calculations we evaluate the expression (14) by taking the approximations

$$\mathbf{v}^{(1)} = \left( \frac{\mu + 2\nu - 4}{2}, \frac{\mu + 2\nu - 3}{2}, \nu - 1 \right) \approx \left( n - 2, n - \frac{3}{2}, \frac{n}{2} - 1 \right) \rightarrow \left( 1, \frac{3}{2}, \frac{1}{2} \right), \quad (15a)$$

$$\mathbf{v}^{(2)} = \left( \frac{\mu + 2\nu - 6}{2}, \frac{\mu + 2\nu - 5}{2}, \nu - 2 \right) \approx \left( n - 3, n - \frac{5}{2}, \frac{n}{2} - 2 \right) \rightarrow \left( 0, \frac{1}{2}, -\frac{1}{2} \right), \quad (15b)$$

as  $n \rightarrow 3$ .

## 2.2 Green function

For the Schwinger-Keldysh or In-in formalism the operators are integrated in different times to then be evaluated at some equal time of observation, which will correspond to the horizon exit of perturbations,  $\tau = \tau_*$  for cosmological applications. From the previous approximations to the truncation of Eq. (14), we can write

$$\begin{aligned} \langle \hat{v}(\mathbf{x}, \tau_1) \hat{v}(\mathbf{y}, \tau_2) \rangle_{\text{reg}} &= A^{(1)} (-ir)^{\alpha(1)} \\ &+ \frac{A^{(1)}}{2} (-i)^{\alpha(1)} r^{\alpha(1)+2} \left[ \frac{1}{(r + \omega_+)^2} + \frac{1}{(r - \omega_-)^2} \right], \end{aligned} \quad (16a)$$

$$A^{(1)}(\mu, \nu) = \frac{\Omega_3}{8(2\pi)^3} \frac{\Gamma(2\nu + \mu)}{\Gamma(2\nu + 1)} \frac{(\nu)_2}{\left(\frac{2\nu + \mu - 2}{2}\right)_2 \left(\frac{2\nu + \mu - 1}{2}\right)_2} (\tau_1 \tau_2)^{\nu-3/2}, \quad (16b)$$

$$\alpha(1) = 4 - \mu - 2\nu, \quad \omega_{\pm} := |\tau_1 \pm \tau_2|, \quad \tau_{1,2} \leq 0. \quad (16c)$$

### 2.3 Equal-time correlator

When interactions are not considered, Eq. (14) is evaluated at equal time, we have

$$\langle \hat{v}(\mathbf{x}, \tau) \hat{v}(\mathbf{y}, \tau) \rangle_{\text{reg}} = A^{(1)} (-ir)^{\alpha(1)} + \frac{A^{(1)}}{2} (-i)^{\alpha(1)} r^{\alpha(1)+2} \left[ \frac{1}{(r + \omega)^2} + \frac{1}{(r - \omega)^2} \right], \quad (17a)$$

$$\omega := |2\tau|, \quad \tau \leq 0. \quad (17b)$$

This result gives a regularized cosmological correlator at tree-level. However, the correlator inherits a regular pole of order two at  $r = \omega$  from the analytic continuation (Ref. [21]) of the hypergeometric function  ${}_1F_2\left(1, \frac{3}{2}, \frac{1}{2}; (2\tau)^2 r^{-2}\right)$  outside of the radius of convergence of the gaussian hypergeometric series<sup>4</sup>. The advantage of this continuation is that it does not have a pole at infinity, this will turn out helpful for the calculations in momentum space.

Furthermore, because the procedure isolates the leading Hadamard singularity of the propagator, it is expected to remain consistent when inserted into higher-order diagrams, where the ultraviolet behavior is governed by the same short-distance structure.

### 2.4 Hadamard structure

The purpose of this subsection is to clarify the regularization procedure used in Eqs. (16)-(17) and to relate it to the short-distance structure of propagators in curved spacetime.

Unlike the point-splitting calculations of Bunch & Davies (Ref. [18]), which were primarily concerned with the coincidence limit of the Green function in order to construct the renormalized stress-energy tensor, the present work focuses on the regulated two-point function at finite separation. Nevertheless, both approaches regulate the same ultraviolet singularity of the propagator, which is known to have the universal Hadamard form.

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<sup>4</sup>The gaussian hypergeometric series has a radius of convergence of  $|z| < 1$  for every  $z \in \mathbb{C}$ . Nonetheless, its analytical continuation can have regular poles at  $z = 1$  or at infinity.

The regularized correlator (17) can be written as

$$\langle \hat{v}(\mathbf{x}, \tau) \hat{v}(\mathbf{y}, \tau) \rangle_{\text{reg}} = A^{(1)} \tau^{2(2\varepsilon-\delta)} \left[ \frac{1}{r^{2+2(2\varepsilon-\delta)}} + \frac{1}{r^{2(2\varepsilon-\delta)}} \left( \frac{1}{(r-2\tau)^2} - \frac{1}{(r+2\tau)^2} \right) \right]. \quad (18a)$$

Thus, by taking the limit  $r \rightarrow 0$ , we have

$$\langle \hat{v}(\mathbf{x}, \tau) \hat{v}(\mathbf{y}, \tau) \rangle_{\text{reg}} \sim \frac{\tau^{2(2\varepsilon-\delta)}}{r^{2+2(2\varepsilon-\delta)}}. \quad (19)$$

This behavior corresponds to the universal Hadamard form of quantum two-point functions in curved spacetime. Exactly the type regulated in point-splitting (Ref. [18]) and discussed in the literature for a de Sitter background (Ref. [19]).

### 3 Regularized propagators

Having obtained the contour-regularized Green functions in position space, Eqs. (16) & (17), we now return to momentum space in order to construct the propagators entering the Schwinger–Keldysh computation of higher-order correlators, in particular the bispectrum relevant for inflationary non-gaussianities [15, 17].

The regularization procedure introduced in the previous section preserves the local Hadamard short-distance structure of the two-point function while rendering its Fourier representation well defined through contour deformation of the spatial kernel. As a result, the propagator admits a regulated momentum-space representation that can be consistently inserted into interacting in-in diagrams.

Unlike subtraction-based renormalization schemes, the present construction retains the ultraviolet structure explicitly at the level of the propagator. This makes it possible to isolate the contribution of short-distance non-Bunch–Davies fluctuations to higher-order correlation functions while maintaining control over the analytic structure of the Schwinger–Keldysh kernel.

The regularized correlator in the position representation is regarded, using the projectors  $|\mathbf{k}\rangle \langle \mathbf{k}|$ ,  $|\mathbf{k}'\rangle \langle \mathbf{k}'|$ , as

$$\langle \hat{v}(x) \hat{v}(y) \rangle_{\text{reg}} = \int \frac{d^n \mathbf{k}}{(2\pi)^n} \frac{d^n \mathbf{k}'}{(2\pi)^n} e^{i(\mathbf{k} \cdot \mathbf{x} - \mathbf{k}' \cdot \mathbf{y})} \langle \hat{v}_{\mathbf{k}}(\tau_1) \hat{v}_{\mathbf{k}'}(\tau_2) \rangle_{\text{reg}}. \quad (20)$$

Nevertheless, instead of working directly with the position projectors  $|\mathbf{x}\rangle \langle \mathbf{x}|$  and  $|\mathbf{y}\rangle \langle \mathbf{y}|$ , it is convenient to exploit the translational and rotational symmetries of the correlator by introducing the variables

$$\mathbf{r} = \mathbf{x} - \mathbf{y}, \quad \boldsymbol{\xi} := \frac{\mathbf{x} + \mathbf{y}}{2}, \quad (21)$$

together with the corresponding projectors  $|\mathbf{r}\rangle \langle \mathbf{r}|$  and  $|\boldsymbol{\xi}\rangle \langle \boldsymbol{\xi}|$ . Here  $\mathbf{r}$  denotes the relative separation, while  $\boldsymbol{\xi}$  plays the role of a center-of-mass coordinate.

Since the regularized correlator in Eq. (14) depends only on the invariant distance  $r = |\mathbf{r}|$ , the integration over the center-of-mass coordinate factorizes, reducing the Fourier representation to an integral over the relative coordinate alone. This is the same symmetry reduction implicitly used in Eq. (11), where the correlator was expressed solely in terms of  $r = |\mathbf{x} - \mathbf{y}|$ .

Furthermore, by eventually reducing the integration to the radial variable  $r$ , rotational invariance is assumed, so that the angular dependence of the relative coordinate is integrated out. It should be stressed the key difference in the contour integration between a different- and an equal-time correlator, is due to the presence of the poles  $\omega_{\pm}$ , which collapses into the same one  $\omega$  for the equal-time Wightman function.

Consequently, the Green function for different poles yields to a causality prescription when applying an off-shell contour deformation in the momenta  $k \rightarrow k - i\epsilon$ . The explicit calculation is derived carefully in the Appendix B, here we just state the main results. On the one hand, through the result of Eq. (B28) the Green function can be written as

$$G_{\text{reg}}^{(\text{rpf})}(\mathbf{k}, \tau_1, \tau_2) := \frac{A^{(1)}\Gamma(s)}{i^{\alpha(1)+s}k^s} - \frac{\pi A^{(1)}}{2i^{\alpha(1)-1}} \sum_{\sigma=\pm} \sigma \theta(\sigma k) \mathcal{F}(\sigma k; \omega_{\sigma}), \quad s = 4 - 2\nu, \quad (22)$$

where the function

$$\mathcal{F}(k; \omega) := e^{i(k-i\epsilon)\omega} (-2\omega - i(k-i\epsilon)\omega^2), \quad (23)$$

encodes the contribution from the double poles, while the Heaviside functions  $\theta(k)$  implement the causal prescription by selecting the appropriate pole depending on the sign of  $k$ . This yields a natural decomposition of the propagator into positive- and negative-frequency sectors.

Furthermore, by taking the limit  $s \approx 1$  at zero order in slow-roll, the regularized propagator in momentum space takes the form

$$G_{\text{reg}}(\mathbf{k}, \tau_1, \tau_2) = \frac{5}{56\pi^2} \left[ \frac{1}{k} + \sum_{\sigma=\pm} \theta(\sigma k) e^{ik\omega_{\sigma}} (-2\omega_{\sigma} - ik\omega_{\sigma}^2) \right], \quad (24)$$

where  $\omega_{\pm} := \tau_1 \pm \tau_2$  and the Heaviside functions  $\theta(\sigma k)$  implement the causal prescription.

This expression exhibits a natural decomposition of the propagator into a local Hadamard contribution and an oscillatory component. The first term,

$$G_{\text{Had}}(k) \sim k^{-1}, \quad (25)$$

corresponds to the universal short-distance singularity of the two-point function. By contrast, the oscillatory contribution behaves as

$$G_{\text{osc}}(k) \sim e^{ik\omega} (\omega + k\omega^2), \quad (26)$$

and is therefore not suppressed in the ultraviolet regime.

Notably, while the Hadamard term encodes the local singular structure of the propagator, it does not carry phase information. The oscillatory contribution, on the other hand, contains the causal phase  $e^{ik\omega}$  and governs the propagation of modes. As a consequence, it becomes dominant in the evaluation of interaction terms, such as those appearing in the bispectrum, where products of propagators and their derivatives probe the full momentum dependence of the Green function.

This decomposition shows that, although the power spectrum is typically controlled by infrared physics, the computation of higher-order correlators involves contributions from subhorizon modes where the oscillatory sector of the propagator plays a leading role.

Contrary to the previous Green function, the equal-time correlator does not exhibit an explicitly causal structure. Nevertheless, such an expression is particularly useful when studying statistical imprints at a fixed observation time, as occurs in inflationary cosmology.

To treat the coincident pole at  $r = \omega$ , the integration contour is deformed in the complex plane as described in Appendix B. Substituting this result into Eq. (B15), together with the Gamma-function contribution obtained previously, the regularized Green function in the relative position frame becomes

$$\boxed{G_{\text{reg}}^{(\text{rpf})}(\mathbf{k}; \tau) = A^{(1)} \left[ \frac{\Gamma(s)}{i^{\alpha(1)+s} k^s} + \frac{\pi}{2 i^{\alpha(1)}} e^{i \operatorname{sgn}(k)(k-i\epsilon)\omega} (2i\omega + |k| \omega^2) \right], \quad (27)$$

$$s = 4 - 2\nu.$$

In the limit  $s \approx 1$ ,  $2\nu \approx 3$  and  $\mu = 3$ , the regularized Green function reduces to

$$G_{\text{reg}}^{(\text{rpf})}(\mathbf{k}; \tau) = A^{(1)} \left[ \frac{i}{k} - \frac{\pi}{2} e^{i \operatorname{sgn}(k)(k-i\epsilon)\omega} (2i\omega + |k| \omega^2) \right], \quad A^{(1)} = \frac{5}{112\pi}. \quad (28)$$

### 3.1 Euclideanized spectral representation

In addition to the Lorentzian contour deformation introduced in the previous subsections, it is useful to consider an auxiliary analytic continuation of the spectral variables which renders the ultraviolet structure manifestly convergent. In particular, we define a Euclideanized representation of the regulated Green function through the formal continuation

$$k \rightarrow ik, \quad \omega \rightarrow i\omega, \quad (29)$$

applied at the level of the momentum-space kernel.

Under this transformation, oscillatory phase factors are mapped into exponentially damped kernels,

$$e^{i \operatorname{sgn}(k)(k-i\epsilon)\omega} \longrightarrow e^{-|k|\omega}, \quad (30)$$

thereby converting UV oscillations into explicit spectral suppression. Similarly, polynomial prefactors become real-valued functions of the Euclideanized variables, ensuring that the resulting kernel admits a manifestly real representation.

Applying this continuation to Eq. (28), the regulated propagator takes the form

$$G_{\text{reg}}^{(\text{E})}(\mathbf{k}; \tau) = A^{(1)} \left[ \frac{1}{k} - \frac{\pi}{2} e^{-|k|\omega} (2\omega + |k|\omega^2) \right], \quad A^{(1)} = \frac{5}{112\pi}. \quad (31)$$

This Euclideanized kernel is not to be interpreted as a physical Wightman function in Lorentzian spacetime, but rather as an auxiliary representation of the same spectral data in which ultraviolet behavior is rendered exponentially convergent. In this sense, the continuation provides a Laplace-type regularization of the oscillatory UV sector, analogous in spirit to heat-kernel or proper-time representations of propagators.

The advantage of this formulation is that the universal short-distance contribution becomes explicitly separated as a scale-invariant term, while state-dependent corrections are encoded in exponentially suppressed contributions. This makes the Hadamard structure of the propagator manifest in a representation where UV convergence is automatic.

Importantly, the Euclideanized kernel is equivalent to the Lorentzian regulated propagator at the level of analytic continuation in the complex momentum plane, provided that one restricts attention to observables reconstructed after inverse continuation to real  $k$  and  $\omega$ . Therefore, physical quantities such as the power spectrum and spectral index are ultimately defined from the Lorentzian theory, while the Euclideanized representation serves as an intermediate regularization device which clarifies the ultraviolet structure of the theory.

### 3.2 Relation to zeta-function regularization and Euclidean continuation

The analytic continuation employed here is conceptually related to zeta-function regularization introduced by Hawking [22]. In that framework the gravitational path integral is evaluated by performing a Wick rotation of the time coordinate,

$$t \rightarrow -it_{\text{E}},$$

such that the Lorentzian Klein–Gordon operator is replaced by an elliptic operator on a Euclidean section of the spacetime. The functional determinants of such operators are then defined through analytic continuation of spectral sums over their eigenvalues.

From a variational approach, a second-order expansion of the action generates the quadratic operator governing the MS field, while the divergent contributions appearing in the correlator are regularized through analytic continuation of their integral representation. Although the present method does not rely explicitly on spectral zeta functions, both procedures share the same underlying principle of assigning finite values to divergent quantities through analytic continuation of the underlying operator kernel.

In Hawking’s construction, the continuation is implemented at the level of the background geometry via Euclidean time, which renders the differential operator elliptic and allows for a spectral decomposition in terms of its eigenmodes. In the present

approach, the continuation is instead implemented at the level of the momentum-space representation of the two-point function. However, as shown in Sec. 3.1, this procedure admits an equivalent interpretation as an analytic continuation of both the momentum variable and the time separation,

$$k \rightarrow ik, \quad \tau \rightarrow i\tau_E,$$

which maps the oscillatory Lorentzian kernel into an exponentially damped Euclideanized representation.

The relation between the two approaches can also be understood from the perspective of kernel methods. In Hawking’s approach the Klein–Gordon operator is diagonalized through a spectral decomposition, leading to a zeta function constructed from its eigenvalues. Equivalently, the same quantities can be expressed through the heat kernel

$$K(x, x', t_E) = \langle x | e^{-t_E \mathcal{O}} | x' \rangle,$$

whose Mellin transform yields the spectral zeta function (Ref. [23]). In the present calculation, the MS equation plays an analogous role: its mode solutions define the kernel of the two-point function through a momentum-space integral representation. The analytic continuation of this representation therefore corresponds to a Laplace-type deformation of the kernel, in which oscillatory Lorentzian phases are mapped into exponentially suppressed Euclidean contributions.

In this sense, the present regularization scheme can be interpreted as a momentum-space realization of the same analytic continuation principle underlying zeta-function and heat-kernel methods, but applied directly at the level of the propagator rather than through an explicit spectral sum of the differential operator.

## 4 Quantum interactions

The regularization procedure introduced in the previous sections was formulated for the Mukhanov–Sasaki variable  $v$ . However, the interaction Hamiltonian is most naturally expressed in terms of the comoving curvature perturbation defined in Eq. (5). Since this relation corresponds to a linear field redefinition,  $\mathcal{R} = z^{-1}v$ , the corresponding two-point functions are related by

$$G_{\mathcal{R}}(\mathbf{k}, \tau_1, \tau_2) = z^{-1}(\tau_1) z^{-1}(\tau_2) G_v(\mathbf{k}, \tau_1, \tau_2), \quad (32)$$

where  $G_v$  denotes the contour-regularized MS propagator of Eq. (24) constructed in the previous sections.

This rescaling does not affect the perturbative slow-roll counting of interaction vertices. Although  $z \sim \sqrt{\varepsilon}$  introduces explicit factors of  $\varepsilon$  in the field redefinition, we follow the standard Maldacena construction of the cubic action and its implementation in the in-in formalism. In this framework, contributions from the field redefinition combine consistently with those generated by the constraint equations and background-dependent vertices. As in the original analysis of [1], these terms reorganize such that potential inverse powers of  $\varepsilon$  cancel in the in-in expansion. As a result,

the bispectrum scales as  $\mathcal{O}(\varepsilon)$ , in agreement with the standard slow-roll consistency of single-field inflation.

Consequently, the contour-regularized propagator for the MS field directly induces a regulated propagator for the curvature perturbation. In particular, both the Hadamard short-distance structure and the non-Bunch-Davies oscillatory contributions are inherited by the  $\mathcal{R}$  correlators through the above rescaling.

The cubic action (See Ref. [1]) contains several interaction terms involving the comoving curvature perturbation, and after removing operators proportional to the linear equations of motion together with subleading slow-roll contributions, the leading interactions take the form

$$\mathcal{H}_{\text{int}} = \lambda_1(\tau) \mathcal{R} \dot{\mathcal{R}}^2 + \lambda_2(\tau) \mathcal{R} (\partial \mathcal{R})^2, \quad (33)$$

with

$$\lambda_1(\tau) \sim \varepsilon(\tau), \quad \lambda_2(\tau) \sim \varepsilon(\tau). \quad (34)$$

#### 4.1 Contribution from $\lambda_1 \mathcal{R} \dot{\mathcal{R}}^2$

The tree-level contribution (See Eq. (E46b) in the Appendix E) to the bispectrum from the interaction term  $\lambda_1(\tau) \mathcal{R} \dot{\mathcal{R}}^2$  is given by

$$\begin{aligned} & \langle \hat{\mathcal{R}}(\mathbf{x}_1, \tau_*) \hat{\mathcal{R}}(\mathbf{x}_2, \tau_*) \hat{\mathcal{R}}(\mathbf{x}_3, \tau_*) \rangle_{\lambda_1} = \\ & -4 \operatorname{Im} \int_{\tau_0}^{\tau_*} d\tau_1 \lambda_1(\tau_1) \sum_{3 \text{ perms}} \int d^3x \frac{d^3k_1}{(2\pi)^3} \frac{d^3k_2}{(2\pi)^3} \frac{d^3k_3}{(2\pi)^3} e^{i\mathbf{k}_1 \cdot (\mathbf{x}_1 - \mathbf{x})} e^{i\mathbf{k}_2 \cdot (\mathbf{x}_2 - \mathbf{x})} \\ & \quad \times e^{i\mathbf{k}_3 \cdot (\mathbf{x}_3 - \mathbf{x})} G(k_1; \tau_*, \tau_1) \partial_{\tau_1} G(k_2; \tau_*, \tau_1) \partial_{\tau_1} G(k_3; \tau_*, \tau_1), \end{aligned} \quad (35)$$

where the sum runs over the three permutations (which external point contracts with the undifferentiated  $\mathcal{R}$ ), and the factor of 4 arises from the in-in prefactor  $(-2)$  and the combinatorial factor 2 from the two identical  $\dot{\mathcal{R}}$  fields.

Then, by the identity  $\int d^n x e^{-i(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3) \cdot \mathbf{x}} = (2\pi)^n \delta(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3)$  we can integrate over position space to obtain

$$\begin{aligned} & \langle \hat{\mathcal{R}}(\mathbf{x}_1, \tau_*) \hat{\mathcal{R}}(\mathbf{x}_2, \tau_*) \hat{\mathcal{R}}(\mathbf{x}_3, \tau_*) \rangle_{\lambda_1} = \\ & -4 \operatorname{Im} \int_{\tau_0}^{\tau_*} d\tau_1 \lambda_1(\tau_1) \sum_{3 \text{ perms}} \int \frac{d^3k_1}{(2\pi)^3} \frac{d^3k_2}{(2\pi)^3} \frac{d^3k_3}{(2\pi)^3} e^{i(\mathbf{k}_1 \cdot \mathbf{x}_1 + \mathbf{k}_2 \cdot \mathbf{x}_2 + \mathbf{k}_3 \cdot \mathbf{x}_3)} \\ & \quad \times (2\pi)^3 \delta(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3) G(k_1; \tau_*, \tau_1) \partial_{\tau_1} G(k_2; \tau_*, \tau_1) \partial_{\tau_1} G(k_3; \tau_*, \tau_1). \end{aligned} \quad (36)$$

Thus, the bispectrum is defined as

$$\begin{aligned} B(k_1, k_2, k_3) &:= \\ & -4 \operatorname{Im} \int_{\tau_0}^{\tau_*} d\tau_1 \lambda_1(\tau_1) \sum_{3 \text{ perms}} G(k_1; \tau_*, \tau_1) \partial_{\tau_1} G(k_2; \tau_*, \tau_1) \partial_{\tau_1} G(k_3; \tau_*, \tau_1), \end{aligned} \quad (37)$$

where the propagators  $G(k; \tau_*, \tau_1)$  are understood to be the contour-regularized Mukhanov–Sasaki Wightman functions constructed in Sec. III in the non–Bunch–Davies Bessel basis.

The relation  $\mathcal{R} = z^{-1}v$  implies that each external curvature perturbation carries a factor of  $z^{-1}(\tau_*)$  evaluated at the final time. These factors therefore affect only the overall normalization of external legs and factorize completely from the time-integrated interaction kernel. In particular, the in-in time evolution is governed solely by the MS propagators  $G_v$  and the interaction Hamiltonian, so that the slow-roll scaling is entirely determined by the structure of  $\lambda_1(\tau)$  and the dynamics of the contour-regularized propagator, independently of the external  $z$ -rescaling.

## 4.2 Non–Bunch–Davies bispectrum

As discussed in the previous subsection, the bispectrum in the in-in formalism is fully determined once the Wightman functions are specified. Therefore, the contour-regularized propagator obtained in momentum space in Eq. (24) can be directly used in Eq. (37) without further modification of the interaction formalism. This provides a consistent evaluation of the tree-level bispectrum in the ultraviolet regime of the theory, where the propagator retains its non–Bunch–Davies oscillatory structure.

A careful treatment of the time integration of Eq. (37) using a deformed time prescription is done in the Appendix C. After taking the imaginary part from Eq. (C42), by means of the identity

$$\frac{1}{ij+1} = e^{-i\pi(j+1)/2}, \quad (38)$$

the time-integrated expression can be written in terms of an explicit oscillatory kernel

$$B_{\text{UV}}(k_1, k_2, k_3) \approx -4\epsilon C^3 \sum_{m=0}^6 \frac{m! A_m(k_1, k_2, k_3)}{K^{m+1}} \sin\left(K\tau_* - \frac{\pi}{2}(m+1)\right), \quad (39)$$

where the oscillatory dependence is governed by the total momentum  $K = k_1 + k_2 + k_3$ .

In the squeezed limit  $k_1 \ll k_2 \simeq k_3 =: k$ , the leading contributions arise from the  $A_5$  and  $A_3$  coefficients. There is no infrared enhancement  $\propto 1/k_1$ . The bispectrum behaves as

$$B_{\text{UV}}^{(\text{sq})}(k_1, k, k) \approx \alpha_{\text{sq}} \left(\frac{5}{56\pi^2}\right)^3 \frac{\epsilon}{k^2} \sin\left(2k\tau_* - \delta_{\text{sq}}\right) + \mathcal{O}\left(\frac{k_1}{k^3}\right), \quad (40)$$

where  $\alpha_{\text{sq}} \simeq 7.65$  and  $\delta_{\text{sq}} \simeq 1.373$  are numerical constants arising from the summed UV polynomial contributions.

Thus, the squeezed configuration remains finite at leading order and does not exhibit the strong infrared enhancement characteristic of non-attractor or explicitly non-adiabatic scenarios. In this sense, the standard single-field consistency behavior is preserved at leading order, although the bispectrum retains oscillatory state-dependent corrections associated with the non–Bunch–Davies initial conditions.

In the equilateral configuration  $k_1 = k_2 = k_3 = k$  (so  $K = 3k$ ), the symmetric coefficients scale as

$$\begin{aligned} A_6 &= -3i k^5, & A_5 &= -18 k^4, & A_4 &= 168i k^3, \\ A_3 &= 72 k^2, & A_2 &= -108i k, & A_1 &= -24. \end{aligned} \quad (41)$$

After inserting into the time-integrated expression and collecting all contributions, every term produces the same overall momentum scaling  $1/k$ . We obtain

$$B_{\text{UV}}^{(\text{eq})}(k, k, k) \approx \alpha_{\text{eq}} \left( \frac{5}{56\pi^2} \right)^3 \frac{\varepsilon}{k} \sin(3k\tau_\star - \delta_{\text{eq}}), \quad (42)$$

where  $\alpha_{\text{eq}} \simeq 2.21 \times 10^4$  and  $\delta_{\text{eq}} \simeq 1.187$  are numerical coefficients obtained from the summed UV contributions.

## 5 Final discussion

In this work we have studied the impact of a non-Bunch-Davies initial state on inflationary correlation functions within a Hadamard-preserving regularization framework. The analysis is based on a Bessel-function mode decomposition of the Mukhanov-Sasaki field, which modifies the analytic structure of the in-in propagator while preserving the standard short-distance Hadamard singularity.

At the level of two-point functions, the regulated propagator contains the usual Minkowski-like Hadamard contribution together with state-dependent oscillatory corrections. These arise from the contour representation of the propagator and are naturally interpreted as a signature of an excited initial state, rather than as ultraviolet renormalization effects.

The main result of this work is the computation of the tree-level bispectrum sourced by cubic interactions. We find a slow-roll suppressed signal of order  $\mathcal{O}(\varepsilon)$  which departs from the standard Maldacena single-field result [1] through oscillatory contributions of the form  $\sin(K\tau_\star)$ . These arise from interference between positive- and negative-frequency components in the non-Bunch-Davies mode basis and encode the finite-time preparation of the initial state.

In contrast with phenomenological approaches to resonant non-Gaussianity, where oscillatory bispectrum templates are typically generated from periodic background interactions or effective ansätze for the in-in time integral [7–10], the present construction derives the oscillatory structure directly from the analytic properties of contour-deformed non-Bunch-Davies propagators. In this sense, the resulting phase dependence should be interpreted as a propagator-level effect associated with the excited initial state, rather than as a resonance induced by an explicitly oscillatory interaction background.

In the squeezed limit, the leading infrared-enhanced contributions cancel after summing the symmetric UV sectors, and the bispectrum remains finite at leading order. Consequently, the result is consistent with the standard single-field attractor

behavior, despite the presence of state-dependent oscillatory corrections associated with the non-Bunch-Davies initial state.

For sub-Hubble scales,  $K|\tau_*| \gg 1$ , the oscillatory sector is dominated by rapidly varying phase contributions associated with the finite-time preparation of the excited initial state. In this regime, the bispectrum can be interpreted as an effective state-dependent short-distance interaction kernel arising from interference effects in the in-in contour representation, rather than as a universal superhorizon attractor observable. Since no infrared enhancement survives in the squeezed configuration after summing the symmetric UV sectors, the signal is dominated by configurations with comparable momenta and is therefore peaked near the equilateral limit while remaining slow-roll suppressed.

Future work could explore connections with other excited-state frameworks, such as  $\alpha$ -vacua and boundary effective field theory descriptions of initial conditions, as well as the mapping of the present result onto standard templates of oscillatory or resonant non-Gaussianity.

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### Competing Interests

The author declares that there are no competing interests.

## Appendix A Hypergeometric integrals

In the particular case of slow-roll, the integral of Eq. (12b) in the BSGM result can be approximated by inverting the relation (2) of the frequency in terms of the space dimension as

$$n = \frac{2(\nu + \delta) - \varepsilon}{1 + \varepsilon} = 2\nu + 2\delta - \varepsilon(1 + 2\nu) + \mathcal{O}(\varepsilon^2, \delta\varepsilon) \approx 2\nu, \quad (\text{A1})$$

where the approximation is taken to order zero in slow-roll. In this way the slow-roll parameters, which depend on the curvature of space-time, are not analytically continue.

By using  $\sin^{2\nu}(\phi) \approx \sin^3(\phi)$ , when  $2\nu \approx n$  and  $n \rightarrow 3$ , we obtain

$$I := \int_0^\pi f(\mathbf{a}, z(\phi)) \sin^3(\phi) d\phi, \quad (\text{A2})$$

where we have defined the function

$$f(\mathbf{a}, z) := {}_2F_1(a_1, a_2; a_3; z), \quad (\text{A3a})$$

$$\mathbf{a} := (a_1, a_2, a_3), \quad a_1 := \frac{\mu + 2\nu}{2}, \quad a_2 := \frac{\mu + 2\nu + 1}{2}, \quad a_3 := \nu + 1. \quad (\text{A3b})$$

Also, by using the notation

$$\mathbf{n} := (n, n, n), \quad \ell(\mathbf{a}) := \frac{a_1 a_2}{a_3}, \quad (\text{A4})$$

the following property of the hypergeometric function

$$\frac{d}{dz} {}_2F_1(a, b; c; z) = \frac{ab}{c} {}_2F_1(a+1, b+1; c+1; z), \quad (\text{A5})$$

can be written as

$$f(\mathbf{a}, z) = \frac{1}{\ell(\mathbf{a} - \mathbf{1})} \frac{d}{dz} f(\mathbf{a} - \mathbf{1}, z). \quad (\text{A6})$$

With this property (A6), the integral (A2) can be solved by parts with the change of variable introduced in Eq. (12c). To see this, let us define an auxiliary function in terms of  $z$  as

$$u(z) := 1 + A(z + B)^2, \quad A := -\frac{a^4}{4b^2 c^2}, \quad B := \frac{b^2 + c^2}{a^2}, \quad dz = -\frac{1}{\sqrt{-A}} \sin(\phi) d\phi,$$

to obtain

$$I = -\sqrt{-A} \left( u(z) \frac{f(\mathbf{a} - \mathbf{1}, z)}{\ell(\mathbf{a} - \mathbf{1})} - \frac{du(z)}{dz} \frac{f(\mathbf{a} - \mathbf{2}, z)}{\ell(\mathbf{a} - \mathbf{1}) \ell(\mathbf{a} - \mathbf{2})} + \frac{d^2 u(z)}{dz^2} \frac{f(\mathbf{a} - \mathbf{3}, z)}{\ell(\mathbf{a} - \mathbf{1}) \ell(\mathbf{a} - \mathbf{2}) \ell(\mathbf{a} - \mathbf{3})} \right) \Big|_{z(0)}^{z(\pi)}. \quad (\text{A7})$$

## Appendix B Contour integral with causal prescription

In this section we compute the integral of Eq. (20) using the contour integration with a causal prescription and the center of mass coordinates from Eqs. (21). The goal is to consistently account for the pole structure of the integrand and obtain a closed analytic expression.

Following this approach, the result of projecting  $\langle \hat{v}(\mathbf{k}, \tau_1) \hat{v}(\mathbf{k}, \tau_2) \rangle$  is

$$\begin{aligned} \langle \hat{v}(\mathbf{k}, \tau_1) \hat{v}(\mathbf{k}, \tau_2) \rangle_{\text{div}} &= \int d^n \mathbf{x} d^n \mathbf{y} e^{-i\mathbf{k} \cdot (\mathbf{x} - \mathbf{y})} \langle \hat{v}(x) \hat{v}(y) \rangle_{\text{reg}} \\ &= \int d^n \boldsymbol{\xi} d^n \mathbf{r} e^{-i\mathbf{k} \cdot \mathbf{r}} \langle \hat{v}(x) \hat{v}(y) \rangle_{\text{reg}} \\ &= \int_{\mathbb{R}^n} d^n \boldsymbol{\xi} \int_{\mathbb{R}^n} d^n \mathbf{r} e^{-i\mathbf{k} \cdot \mathbf{r}} \langle \hat{v}(x) \hat{v}(y) \rangle_{\text{reg}} \\ &\approx (\Omega_n)^2 \int_0^\infty d\xi \xi^{n-1} \int_0^\infty dr r^{n-1} e^{-ikr} \langle \hat{v}(x) \hat{v}(y) \rangle_{\text{reg}}. \quad (\text{B8}) \end{aligned}$$

The first integral in Eq. (B8) is divergent and can be identified as an infinite volume from the center of mass frame. To solve it, dimensional regularization needs to be applied. Thus, by using the approximation of Eq. (A1) we can define the volume integral

$$V_{\text{div}}(\nu, l) := \int_0^l d\xi \xi^{2\nu-1}, \quad (\text{B9})$$

and, as  $l \rightarrow \infty$ , the volume turns out divergent. This infinite volume can be regularized via the  $\mu$  function of Ref. [24],

$$\mu(s) := \int_1^\infty dz z^{-s} \stackrel{(\text{a.c.})}{=} \frac{1}{s-1}, \quad s \in \mathbb{C} \setminus \{1\}. \quad (\text{B10})$$

First, we perform a change of variable in terms of  $z = \sqrt{\xi^2 + 1}$ ;  $z \, dz = \xi \, d\xi$ , to write

$$V_{\text{div}}(\nu, l) = \int_1^{z(l)} dz \, z (z^2 - 1)^{\nu-1} \approx \int_1^{z(l)} dz \, z^{2\nu-1}, \quad z(l) = \sqrt{l^2 + 1}.$$

By taking  $l \rightarrow \infty$ , then  $z(l) \rightarrow \infty$ , so the integral can be written as

$$\lim_{l \rightarrow \infty} V_{\text{div}}(\nu, l) = \int_1^\infty dz \, z^{2\nu-1} \stackrel{\text{(a.c.)}}{=} \frac{1}{-2\nu}. \quad (\text{B11})$$

Therefore, we can define the regularized volume, via the analytic continuation (B11), as

$$V_{\text{reg}}(\nu) := -\frac{1}{2\nu}. \quad (\text{B12})$$

The second integral in Eq. (B8) can be identified with the position correlator, as well as with a *Green function*, both in the relative position frame (rpf). This quantity can in principle be divergent. Let it be defined as

$$G_{\text{div}}^{(\text{rpf})}(\mathbf{k}; \tau_1, \tau_2) := \int_0^\infty dr \, r^{n-1} e^{-ikr} \langle \hat{v}(\mathbf{x}, \tau_1) \hat{v}(\mathbf{y}, \tau_2) \rangle_{\text{reg}}, \quad (\text{B13})$$

where  $k = |\mathbf{k}|$  and the integral is performed after substituting the regularized expression

$$\langle \hat{v}(\mathbf{x}, \tau_1) \hat{v}(\mathbf{y}, \tau_2) \rangle_{\text{reg}} = A^{(1)} (-ir)^{\alpha(1)} + \frac{A^{(1)}}{2} (-i)^{\alpha(1)} r^{\alpha(1)+2} \left[ \frac{1}{(r + \omega_+)^2} + \frac{1}{(r - \omega_-)^2} \right]. \quad (\text{B14})$$

By expanding the sum of the integrand and using the approximations of Eqs. (15) & (A1), when  $n \rightarrow 3$ , it can be found that

$$\begin{aligned} G_{\text{div}}^{(\text{rpf})}(\mathbf{k}; \tau_1, \tau_2) &\approx \frac{A^{(1)}}{i^{\alpha(1)}} \int_0^\infty dr \, r^{n+\alpha(1)-1} e^{-ikr} \\ &\quad - \frac{A^{(1)}}{2 i^{\alpha(1)}} \int_0^\infty dr \, r^{n+\alpha(1)+1} e^{-ikr} \left[ \frac{1}{(r + \omega_+)^2} + \frac{1}{(r - \omega_-)^2} \right] \\ &\approx \frac{A^{(1)}}{i^{\alpha(1)+s} k^s} \int_0^\infty d(ikr) \, (ikr)^{s-1} e^{-(ikr)} \\ &\quad - \frac{A^{(1)}}{2 i^{\alpha(1)}} \int_0^\infty dr \, r^2 e^{-ikr} \left[ \frac{1}{(r + \omega_+)^2} + \frac{1}{(r - \omega_-)^2} \right], \quad (\text{B15}) \\ s &:= n + \alpha(1). \end{aligned}$$

Also, in order to obtain expression (B15), the following criteria has been applied:

1. In the first integral, one can see that  $\alpha(1) = -2n - (n+1)\varepsilon + 2\delta + 4 \approx -2$ , such that the exponent  $n + \alpha(1) - 1 \approx 0$ , when  $n \rightarrow 3$ , at order zero in slow-roll. In spite of that, we take  $\alpha(1) = -2 - 4\varepsilon + 2\delta$  at order one in slow-roll, on the grounds that in this fashion the integral relates the space dimension  $n = 3$  with the slow-roll parameters. Thus, in this way the integral captures the physics of the quasi-de Sitter space-time <sup>5</sup>.
2. However, in the second integral the approximation  $\alpha(1) \approx -2$  at order zero in slow-roll is taken, since in this manner the pole can be tamed more easily. In addition, on account of the existence of the pole at  $r = \omega$ , the estimation that this pole carries the significant part of the divergent contribution is a rational approximation since the first integral already carries the non-divergent contribution.

In view of this criteria, the first integral in Eq. (B15) can be calculated by a Gamma function. On the other hand, the second integral requires a more cautious treatment because of the regular poles at  $r = \omega_{\pm}$ . To treat these poles an integration along a semi-circle of radius  $R$  must be performed. The integrand is defined as

$$g(z) := e^{-ikz} z^2 \left[ \frac{1}{(z + \omega_+)^2} + \frac{1}{(z - \omega_-)^2} \right]. \quad (\text{B16})$$

The integral of the function  $g(z)$  in Eq. (B16) along the whole contour needs to be deformed properly within the off-shell prescription in order to converge properly. On these grounds, we define the function as

$$g(z) = e^{-i(k-i\epsilon)z} z^2 \left[ \frac{1}{(z + \omega_+)^2} + \frac{1}{(z - \omega_-)^2} \right], \quad \epsilon > 0, \quad (\text{B17})$$

where the shift  $k \rightarrow k - i\epsilon$  implements the standard causal prescription. This ensures exponential damping of the integrand in the appropriate half-plane and allows for a well-defined contour deformation.

While integral of interest is

$$J := \int_0^\infty dx g(x), \quad (\text{B18})$$

to treat both poles symmetrically, we extend the integration domain to the full real axis

$$J = \frac{1}{2} \int_{-\infty}^\infty dx g(x). \quad (\text{B19})$$

Because the integrand has double poles at

$$z = \omega_-, \quad z = -\omega_+, \quad (\text{B20})$$

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<sup>5</sup>The relevance of this parameters stems from the fact that the quantized field  $v$  arises from a perturbation in quasi-de Sitter space, such that the slow-roll parameters measure the deviation from quasi-de Sitter to the de Sitter background.

then the causal prescription shifts them infinitesimally off the real axis as

$$\omega_- \rightarrow \omega_- - i0, \quad (\text{B21})$$

$$-\omega_+ \rightarrow -\omega_+ + i0. \quad (\text{B22})$$

Thus, we have:

- The pole at  $z = \omega_-$  lies just below the real axis.
- The pole at  $z = -\omega_+$  lies just above the real axis.

Because of the poles, the contour is closed depending on the sign of  $k$ :

- For  $k > 0$ , close in the upper half-plane.
- For  $k < 0$ , close in the lower half-plane.

Since the poles are of second order, the residue at  $z = a$  is given by

$$\text{Res}_{z=a} g(z) = \frac{d}{dz} \left[ e^{-i(k-i\epsilon)z} z^2 \right] \Big|_{z=a}. \quad (\text{B23})$$

A direct computation yields

$$\frac{d}{dz} \left( e^{-i(k-i\epsilon)z} z^2 \right) = e^{-i(k-i\epsilon)z} (2z - i(k-i\epsilon)z^2). \quad (\text{B24})$$

Evaluating at each pole, it leads to

**Pole at  $z = \omega_-$ :**

$$\text{Res}_{\omega_-} = e^{-i(k-i\epsilon)\omega_-} (2\omega_- - i(k-i\epsilon)\omega_-^2). \quad (\text{B25})$$

**Pole at  $z = -\omega_+$ :**

$$\text{Res}_{-\omega_+} = e^{i(k-i\epsilon)\omega_+} (-2\omega_+ - i(k-i\epsilon)\omega_+^2). \quad (\text{B26})$$

Applying Cauchy's theorem, we obtain

$$\int_{-\infty}^{\infty} dx g(x) = 2\pi i [\theta(k) \text{Res}_{-\omega_+} - \theta(-k) \text{Res}_{\omega_-}], \quad (\text{B27})$$

where  $\theta(k)$  is the Heaviside function.

Therefore, the original integral becomes

$$\begin{aligned} J &= \int_0^{\infty} dx g(x) \\ &= \pi i \left[ \theta(k) e^{i(k-i\epsilon)\omega_+} (-2\omega_+ - i(k-i\epsilon)\omega_+^2) \right. \\ &\quad \left. - \theta(-k) e^{-i(k-i\epsilon)\omega_-} (2\omega_- - i(k-i\epsilon)\omega_-^2) \right]. \end{aligned}$$

Substituting into Eq. (B15), the divergent part of the propagator becomes

$$G_{\text{div}}^{(\text{rpf})}(\mathbf{k}, \tau) = \frac{A^{(1)}\Gamma(s)}{i^{\alpha(1)+s}k^s} - \frac{A^{(1)}}{2i^{\alpha(1)}} J, \quad (\text{B28})$$

with  $J$  given by Eq. (B28).

This result exhibits several important features:

- The  $i\epsilon$  prescription enforces causality by selecting the relevant pole depending on the sign of  $k$ .
- The propagator splits into positive- and negative-frequency contributions.
- The pole contributions encode the local singular structure, while the exponential factors reflect propagation along characteristic scales  $\omega_{\pm}$ .

This representation provides a fully analytic and causal evaluation of the integral, consistent with the expected structure of quantum propagators.

## B.1 Coincident pole limit

A particularly relevant case corresponds to the coincident pole configuration

$$\omega_+ = \omega_- = \omega. \quad (\text{B29})$$

In this limit, Eq. (B28) simplifies considerably. The integral becomes

$$J = \pi i \left[ \theta(k) e^{i(k-i\epsilon)\omega} (-2\omega - i(k-i\epsilon)\omega^2) - \theta(-k) e^{-i(k-i\epsilon)\omega} (2\omega - i(k-i\epsilon)\omega^2) \right]. \quad (\text{B30})$$

Using the identity

$$\theta(k) - \theta(-k) = \text{sgn}(k), \quad (\text{B31})$$

together with

$$|k| = \theta(k)k - \theta(-k)k, \quad (\text{B32})$$

the previous expression may be written in the compact form

$$J = -\pi e^{i \text{sgn}(k)(k-i\epsilon)\omega} [2i\omega + |k|\omega^2]. \quad (\text{B33})$$

Therefore, the divergent part of the propagator becomes

$$G_{\text{div}}^{(\text{rpf})}(\mathbf{k}, \tau) = \frac{A^{(1)}\Gamma(s)}{i^{\alpha(1)+s}k^s} + \frac{\pi A^{(1)}}{2i^{\alpha(1)}} e^{i \text{sgn}(k)(k-i\epsilon)\omega} (2i\omega + |k|\omega^2). \quad (\text{B34})$$

## Appendix C Time integral

In this section we compute the time integral of Eq. (37) via our non-Bunch-Davies vacuum prescription. First, let us take the time derivative of the regulated propagator

with respect to  $\tau_1$ . This derivative acts only on the oscillatory sector of the propagator, since the Hadamard contribution is time-independent in the equal-time UV approximation. The resulting expression is given by

$$\partial_{\tau_1} G(k; \tau_*, \tau_1) = C \sum_{\sigma=\pm} \theta(\sigma k) e^{ik\omega_\sigma} (k^2 \omega_\sigma^2 - 4ik\omega_\sigma - 2), \quad C := \frac{5}{56\pi^2}. \quad (\text{C35})$$

Substituting Eqs. (24)&(C35) into Eq. (37), we obtain

$$B_{\text{UV}}(k_1, k_2, k_3) = -4C^3 \text{Im} \sum_{\text{perms}} \sum_{\sigma_i=\pm} \int_{\tau_0}^{\tau_*} d\tau_1 \lambda_1(\tau_1) \theta(\sigma_1 k_1) \theta(\sigma_2 k_2) \theta(\sigma_3 k_3) \\ \times e^{i\Phi(\tau_1)} \mathcal{P}_1(\tau_1) \mathcal{Q}_2(\tau_1) \mathcal{Q}_3(\tau_1), \quad (\text{C36})$$

where the phase is given by

$$\Phi(\tau_1) = k_1 \omega_{\sigma_1} + k_2 \omega_{\sigma_2} + k_3 \omega_{\sigma_3} = (k_1 + k_2 + k_3) \tau_* + (\sigma_1 k_1 + \sigma_2 k_2 + \sigma_3 k_3) \tau_1, \quad (\text{C37})$$

and the polynomial factors read

$$\mathcal{P}_i(\tau_1) = -2\omega_{\sigma_i} - ik_i \omega_{\sigma_i}^2, \quad (\text{C38a})$$

$$\mathcal{Q}_i(\tau_1) = k_i^2 \omega_{\sigma_i}^2 - 4ik_i \omega_{\sigma_i} - 2. \quad (\text{C38b})$$

We further extend the isotropic, as well as the UV approximations, by taking  $k_i = |\mathbf{k}_i|$ . In this regime, for physical momenta  $k_i > 0$ , only the  $\sigma_i = +1$  sector contributes.

This leads to the simplified bispectrum expression

$$B_{\text{UV}}(k_1, k_2, k_3) \approx -4C^3 \text{Im} \sum_{\text{perms}} \int_{\tau_0}^{\tau_*} d\tau_1 \lambda_1(\tau_1) e^{iK(\tau_* + \tau_1)} \mathcal{P}_1(\tau_1) \mathcal{Q}_2(\tau_1) \mathcal{Q}_3(\tau_1), \quad (\text{C39a})$$

$$K := k_1 + k_2 + k_3. \quad (\text{C39b})$$

The interaction  $\lambda_1(\tau_1) \mathcal{R} \dot{\mathcal{R}}^2$  depends only weakly on time in the slow-roll regime. To leading order it can therefore be treated as approximately constant and evaluated at horizon crossing,

$$\lambda_1(\tau_1) \simeq \lambda_{1*} \sim \varepsilon,$$

allowing it to be taken outside the time integral.

The time integrand is then a purely polynomial function of

$$u := \tau_* + \tau_1, \quad \omega_+ = u. \quad (\text{C40})$$

The full UV kernel product generates a polynomial up to sixth order in  $u$

$$\mathcal{P}_1(\tau_1)\mathcal{Q}_2(\tau_1)\mathcal{Q}_3(\tau_1) = \sum_{m=0}^6 a_m^{(1|23)} u^m,$$

and after summing over permutations this defines the fully symmetric coefficients

$$A_m(k_1, k_2, k_3) := \sum_{\text{perms}} a_m^{(i|jk)}. \quad (\text{C41})$$

The in-in prescription selects the interacting vacuum via the standard  $i\epsilon$  deformation,

$$\tau_0 \rightarrow -\infty(1 - i\epsilon), \quad e^{iKu_0} \rightarrow 0,$$

so that only the upper boundary contributes. The time integral, Eq. (C39), can therefore be expressed as

$$B_{\text{UV}}(k_1, k_2, k_3) \approx -4\epsilon C^3 \text{Im} \left[ e^{iK\tau_*} \sum_{m=0}^6 A_m(k_1, k_2, k_3) \frac{m!}{(iK)^{m+1}} \right], \quad (\text{C42})$$

where the symmetric coefficients  $A_m$  are functions of the invariants

$$K = k_1 + k_2 + k_3, \quad s_2 := k_1 k_2 + k_1 k_3 + k_2 k_3, \quad s_3 := k_1 k_2 k_3, \quad (\text{C43})$$

and are explicitly given in Appendix (D).

## Appendix D Permutation Coefficients

In the UV approximation ( $\sigma_i = +1$ ), after expansion, the non-vanishing coefficients for fixed leg ordering (1|23) are

$$a_6^{(1|23)} = -ik_1 k_2^2 k_3^2, \quad (\text{D44a})$$

$$a_5^{(1|23)} = -2k_2^2 k_3^2 - 4k_1(k_2 k_3^2 + k_3 k_2^2), \quad (\text{D44b})$$

$$a_4^{(1|23)} = 2ik_1(k_2^2 + k_3^2 + 8k_2 k_3) + 8i(k_2 k_3^2 + k_3 k_2^2), \quad (\text{D44c})$$

$$a_3^{(1|23)} = 4(k_2^2 + k_3^2 + 8k_2 k_3) + 8k_1(k_2 + k_3), \quad (\text{D44d})$$

$$a_2^{(1|23)} = -4ik_1 - 16i(k_2 + k_3), \quad (\text{D44e})$$

$$a_1^{(1|23)} = -8, \quad (\text{D44f})$$

$$a_0^{(1|23)} = 0. \quad (\text{D44g})$$

The final symmetric coefficients  $A_m(k_1, k_2, k_3)$  are

$$A_6 = -i s_3 s_2, \quad (\text{D45a})$$

$$A_5 = -2 \sum_{i < j} k_i^2 k_j^2 - 4s_3 K, \quad (\text{D45b})$$

$$A_4 = i \left[ 10 \sum_{i \neq j} k_i^2 k_j + 48s_3 \right], \quad (\text{D45c})$$

$$A_3 = 8(k_1^2 + k_2^2 + k_3^2) + 48s_2, \quad (\text{D45d})$$

$$A_2 = -36i K, \quad (\text{D45e})$$

$$A_1 = -24, \quad (\text{D45f})$$

$$A_0 = 0, \quad (\text{D45g})$$

where

$$K = k_1 + k_2 + k_3, \quad s_2 = k_1 k_2 + k_1 k_3 + k_2 k_3, \quad s_3 = k_1 k_2 k_3.$$

These coefficients fully determine the UV contribution to the bispectrum and encode all momentum dependence in a symmetric polynomial basis.

## Appendix E In-in formalism

In the In-in or Schwinger-Keldysh formalism, the correlators of an observable  $W(t)$  in the interaction picture are calculated as

$$\langle W(t_\star) \rangle_0 = \langle W_{\text{int}}(t_\star) \rangle, \quad (\text{E46a})$$

$$\langle W(t_\star) \rangle_1 = -2 \text{Im} \left\{ \int_{t_0^-}^{t_\star} dt_1 \langle H_{\text{int}}(t_1) W_{\text{int}}(t_\star) \rangle \right\}, \quad (\text{E46b})$$

$$\begin{aligned} \langle W(t_\star) \rangle_2 = & \int_{t_0^-}^{t_\star} dt_2 \int_{t_0^+}^{t_\star} dt_1 \langle H_{\text{int}}(t_2) W_{\text{int}}(t_\star) H_{\text{int}}(t_1) \rangle \\ & - 2 \text{Re} \left\{ \int_{t_0^-}^{t_\star} dt_2 \int_{t_0^-}^{t_\star} dt_1 \langle H_{\text{int}}(t_1) H_{\text{int}}(t_2) W_{\text{int}}(t_\star) \rangle \right\}, \end{aligned} \quad (\text{E46c})$$

for the non-interacting, tree-level and one-loop interactions, respectively.

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